Autonomous satellite control for observation – machine learning

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Objective

- Observe the target for as many frames as possible from as close as possible
- Difficulties: clouds can block geometrically optimal pointing maneuver
- If we do “geometrically optimal dumb” we might need more orbits to get a picture we want which is suboptimal if we want an observation ASAP (this is relevant if the observation target can move far in 96 minutes)

Machine learning solution

- Depending on weather clouds may have gaps
- If the satellite “sees” the gaps he can anticipate where to do a pointing manouver to see “under the cloud” at an angle
- Complex optimization objective, clouds move

State space:

- Orbit location r_sat(t) deterministic, given by orbit
- Satellite attitude orientation ao_sat(t), this is controllable
- Clouds location r_cloud(t) “random” variable
- Target location r_target, fixed
- Air drag: negligible

Action space:

- Torque, reaction wheel -> change in ao_sat(t)
- “Passive observation” -> always
- “Active observation” -> take picture and downlink to mission control

ML Objectives

- Objective:
  - clear (no cloud occlusion) picture
  - High resolution (close) picture
- Secondary objectives:
  - No downlinks for pictures too far or occluded
  - No random torque actions (energy wasted)

Reward function v1: r1(s, a)

- All rewards per timestep, only primary objectives for v1
- No Picture: -100
- Picture without visible target: -100
- Picture beyond resolution threshold distance: -100
- Optimal distance (nadir):= d_op, threshold distance := d_th
- Picture inside threshold distance:
  - R1 = -100 * |r_t – r_sat| * 1/(d_th-d_op)